Let us consider a system which undergoes a change of state from state (1) to state (2) at constant temperature.

$G_2 - G_1 = (H_2 - H_1) - T(S_2 - S_1)$

$\Delta G = \Delta H - T\,\Delta S$ ------(7.36)

Now let us consider how ΔG is related to reaction spontaneity.

We know that

$\Delta S_{total} = \Delta S_{sys} + \Delta S_{surr}$

For a reversible process (equilibrium), the change in entropy of universe is zero. $\Delta S_{total} = 0$ [$\because \Delta S_{sys} = -\Delta S_{surr}$]

Similarly, for an equilibrium process ΔG=0

For spontaneous process, $\Delta S_{total} > 0$

$\Delta S_{sys} + \Delta S_{surr} > 0$

$\Delta S_{sys} + \frac{dq_{surr}}{T} > 0$

$\Delta S_{sys} - \frac{\Delta H_{sys}}{T} > 0$

$T\Delta S_{sys} - \Delta H_{sys} > 0$

$-(\Delta H_{sys} - T\Delta S_{sys}) > 0$

$-\Delta G > 0$

hence for a spontaneous process,

$\Delta G < 0$

i.e. $\Delta H - T\,\Delta S < 0$ ------(7.37)

ΔH_{sys} is the enthalpy change of a reaction, $T\Delta S_{sys}$ is the energy which is not available to do useful work. So ΔG is the net energy available to do useful work and is thus a measure of the 'free energy'. For this reason, it is also known as the free energy of the reaction. For non spontaneous process, ΔG > 0

Gibbs free energy and the net work done by the system:

For any system at constant pressure and temperature

$\Delta G = \Delta H - T\,\Delta S$ ---- (7.36)

We know that,

$\Delta H = \Delta U + P\Delta V$

$\therefore\ \Delta G = \Delta U + P\Delta V - T\Delta S$

from first law of thermodynamics

if work is done by the system

$\Delta U = q - w$

from second law of thermodynamics

$\Delta S = \frac{q}{T}$

$\Delta G = q - w + P\Delta V - T\left(\frac{q}{T}\right)$

$\Delta G = -w + P\Delta V$

$-\Delta G = w - P\Delta V$ .....(7.38)

Here, $-P\Delta V$ represents the work done due to expansion against a constant external pressure. Therefore, it is clear that the decrease in free energy ($-\Delta G$) accompanying a process taking place at constant temperature and pressure is equal to the maximum work obtainable from the system other than the work of expansion.

7.11.1 Criteria for spontaneity of a process

The spontaneity of any process depends on three different factors.

- If the enthalpy change of a process is negative, then the process is exothermic and may be spontaneous. (ΔH is negative)
- If the entropy change of a process is positive, then the process may occur spontaneously. (ΔS is positive)

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